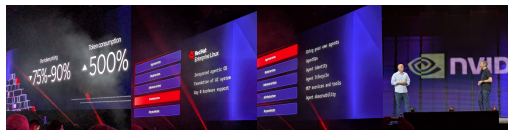


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AI agents, open source, hybrid cloud, virtualization costs, regulatory environment, digital sovereignty, OpenShift, Kubernetes, generative AI, inference services, model services, agent services, token economics, network automation, customer centricity.



SPEAKERS

Speaker 10, Speaker 1, Speaker 7, Speaker 9, Speaker 2, Speaker 6, Speaker 4, Speaker 8, Speaker 5, Speaker 3

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Speaker 1 00:10

Look at this crap. Every one of you is responsible for something that cannot break, cannot go down. Cannot be wrong, whether it's millions of financial transactions being processed every hour or healthcare networks that physicians depend on for life affecting decisions, or the millions of consumers served by telecommunications networks. That's just a normal week for the people in this room. And here's what I think about a lot. The outcomes get all the attention, the deployment measures, the SLAs, the uptime measurements, what doesn't get seen is the craft behind them that makes it possible, the person that designed the architecture from scratch, the person that migrated hundreds of virtual machines invisibly to the business. The person that at 2am in the morning figures out the cause of an outage that wasn't in the run book that takes real expertise, real creativity, and yet most of the world never sees it? I see it, and I think it's important to address that before we talk about anything else today. Now, the question for you hands up and be honest with me on this one, how many of you are managing an environment today that is more complex than what you were managing two years ago. Put your hands up. Okay. Now, keep your hands up if your budget has grown at the same rate as that complexity. Yeah, there aren't a lot of hands left. Now think about in addition to this, how many of you are being asked to run an AI initiative while also managing a legacy system that's absolutely critical to your business. So these constraints, these pressures, the new initiative pressure, the organizational friction, the legacy constraints, all happening at once. That is the daily reality for the people in this room. That's the combination is the villain in the room. Think about your reality. Your virtualization costs are out significantly. Your regulatory environment is expanding beyond your compliance team's ability to keep up your board is probably pushing for ROI on AI projects at this point that might not match your enterprise reality, and yet, your people, your talented, hard working people, are being asked to modernize and secure and operate and innovate on the same timelines with the same budget, that gap, that chasm

between the complexity of your environment and the resources you have to work with it. And the speed at which your business needs to compete, that is the real story of enterprise. It right now, not just AI. And if we're not talking about that chasm, then we're not talking about your reality. Now I've actually seen these pressures before that lead to a question at practically every customer that I talk to, what do I build on? Not what vendor, not what Cloud, not what model, what is the foundation that lets me compete at the speed that my business needs, while not breaking what absolutely cannot be broken my business. I've seen this before, twice, actually, both times in ways that fundamentally reshaped the industry, the first time was in the shift from proprietary hardware and proprietary operating systems to open infrastructure to that question, the what do I build On question? The answer turned out to be Linux. Red Hat brought Linux to the enterprise. With Red Hat, enterprise, Linux and REL became a standard practically every industry. Everything that followed was built on Linux, cloud, IoT, mobile. Think about today. Every AI model, every AI agent, runs on Linux. That foundation that we built together is now the reality of what everything we depend on today is running. Now. The second time this came up was in the shift to cloud native So containers, Kubernetes, micro services. Now at Red Hat, we didn't invent Docker. We didn't invent Kubernetes. We brought Kubernetes to the enterprise through OpenShift, and now that is a standard that 1000s of customers depend on for their hybrid cloud. We are at the third inflection point right now with AI and Red Hat will bring AI to the enterprise to make that an open standard. And I think the pattern holds from the last two the organizations that will have the most success in this environment, they are not necessarily going to be the ones that spend the most or the ones that move the fastest. They will be the ones that build on the right foundation, virtual machines, containers, AI agents. They are not competing with each other. They are converging. And the answer to this is not going to be one cloud or one vendor or one model, it will be the right platform with a broad ecosystem that supports it, and I'm biased, but I think the organizations that will have the most success will pick a platform with an open foundation below. Now, speaking of AI, let's take a minute and talk about our AI journey at Red Hat, because we're going through the same inflection point at the same time as all of you. As you know, Red Hat is an open source company, so what we build, we share. That is how open source works, and at this point, we have deployed generative AI to every organization in the company, each organization is looking for ways to either drive more efficiency for them or create more value for you customers. For example, you might have heard about or used as

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Speaker 2 07:57

red, which is our

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Speaker 1 07:57

interactive chat bot it runs on our customer support. Now, this chatbot has been trained on over two decades of support information, knowledge base, support capabilities within Red Hat, and this is a case where hopefully this is more value for you when you're operating in the known of support, but it is also more efficient for us and Saves us millions of dollars a year. Now in this journey, we've had plenty of successes, we have also had plenty of failures, and that's easy to do when the technology is changing practically weekly now, but those learnings are going actively back into the platforms that we provide you, everything you're going to hear about at Summit that our confidence, our belief that we can be the right foundation for your

journey. It's not just a market position, it's what we are actually doing in Red Hat. Now, about a year ago, we moved from the world of chatbots to building AI agents. Specifically, we wanted to build an agentic system that could go through a massive body of internal knowledge and return something genuinely useful, not a summary, but an actual answer. We started where many of you probably start with, Frontier models, the best, most capable models that we could work with. We wanted to understand what was possible before we decided how to build this right. And it worked. After we had about 10 agents working together, it was clear that this system worked and would continue to work. So we did what was very natural to Red Hat. We started to optimize. We went through layer by layer, figuring out which of these models could be replaced by open weight models running on bllm, on infrastructure we control. First, we actually rebuilt the document search capabilities. We found out that small models excel in the correctly formatted documents of finding the right answers. So we moved document search. Next came hallucination detection. Our 10 initial agents has now scaled to almost 200 and you will deal with hallucinations. Small models were an excellent way for us to do that, double checking and hold hallucinations a day. We moved from hallucinations to safety management to controlling what we would answer and commit small models were incredible there. Lastly, we move to planning the most complicated step, where an agent has to look at all the other agents it could talk to, and figure out what questions to ask to get that genuinely useful answer, layer by layer, we went through the entire application to figure out what we could switch from the frontier models to open source models running on our infrastructure and technology today, 85% 5% of the calls running on our deep research agent system called Open Source open weight models running on Red Hat infrastructure. And here's what surprised us, of course, this was more efficient, but our results actually got better, and it wasn't because frontier models weren't the right place to start, but when you own the model and you own the infrastructure, you can optimize in a way that a frontier model that its general purpose could never anticipate. You anticipated. The heart of our system today is running on Nemo, Tron, super, nematron, nano and IBM granite. This is all running on Red Hat infrastructure, and now we are at an efficiency point a cost, a unit cost perspective, that we are able to start rolling this out to the entire company. What we built not a demo. This wasn't a pilot. This is fundamentally changing how we operate at Red Hat and our learnings go directly back into the platforms that we provide for you all. That is how open source works. But beyond the technical proof, beyond having a system that works the power of hybrid AI, seeing what smaller open weight models can do, here is what we learned about the process, the actual work. And I want to be direct on this, to these software developers in the room. You are not getting replaced by AI, but where you spend your time and energy really drastically figuring out how to build and shape evaluation for these AI created systems figuring out how to build the next generation of continuous integration systems in a world where anyone can contribute, Figuring out how to create the next generation testing frameworks that maintain some stability, the humans that build the rails that AI runs on. That is where your craft is going to, the managers in the room you are about to be tested on your ability to delegate now AI agents, they are going to expand the capabilities and capacity of your team, in some cases, drastically. They're also going to pressure test every process that you have, and in your case, as a manager, they will pressure test your ability to understand the technology, truly understand your processes, and be able to decompose complicated work, to be able To delegate to both humans and agents. Your ability to do this will largely define how well you can lead in this environment and to every other person in every role in the company. You now have no barriers to contribute your knowledge into a system like this. At this point, we have had content knowledge contributed into our deep research agent from every organization in the company, but beyond knowledge, beyond content, we have had code contributed from every organization in the company, from legal, from inside sales, from operations, from teams that have never written a line of production. Code in their careers have been able to positively impact our agentic system when you couple

AI and its ability to amplify what we can do with a well engineered system, I promise you it will Change what you can build as a company. AI isn't optional at this point, whether it brings you energy or whether it creates real uncertainty. And I think for many of us, it is a combination, a mixture of both. It can't be ignored. But there also isn't a single path. Some of you might be ready to run AI or are in production. Some of you might be just trying to figure out what is hype and what is reality at this point, and some of you might be just trying to get your infrastructure in order to even make this a reality, all of that is legitimate, what I would caution against are the extremes, either moving so slow that when you Start you can never catch up, or moving so fast that you build on a foundation that you cannot address what comes next, because I promise you, something will come next in this environment. So here's what I can promise you, everything you're going to hear about at Summit is what Red Hat is actually doing. Will not be theory, won't be roadmap slides. This will be production experience. I've genuinely never been more excited about the potential of the application of this technology, and also that the timing is perfect for us to figure out this journey together. Let's get to it. Welcome to Red Hat summit 2026 please welcome

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Speaker 3 18:19

Red Hat Senior Vice President and Chief red of the and Chief Product Officer. Ashish badani,

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Speaker 4 18:33

thank you, Matt. Let's talk about something that we've heard from hundreds of customers, perhaps many of you in this room for the last 12 to 18 months, there is a virtualization cost crisis. That's only part of the infrastructure challenge that you're facing. The deeper problem is this, your infrastructure layer needs to increasingly do more than you plan for niche to support virtual machines, for your legacy of mission critical application. Need to support containers, cloud native applications for the workloads that your developers are building today and for tomorrow, and for AI, as you heard from Matt, needs to support a future where you have GPU intensive intraceby workloads, three fundamentally different workload types. Now most platforms will ask you to independently manage them, three different ways, three different operation models and three different cost structures. We feel that's an unsustainable ask for any chain where there was one platform with one operational model and one consistent security posture. That is exactly what OpenShift delivers, and it's different from every other platform in the market today, virtual machines, containers and AI workloads, all running side by side, manage the same way, operate the same way, secure the same way, providing that platform though just one part of the answer, as you all well know, migrating that modulation environment, it's not a weekend project, it's complex, touches hundreds of applications, carries real business risk. We know this because we have done this with some of our largest customers at scale, and so have a partners. We know something else too. The hardest part that work is not the technology. It's never been the technology. It's the people, the engineers, who have to keep the old environment running at full availability while simultaneously building the new one, all without additional headcap and without missing a single SLA in the process. Now that is a big ask. That's exactly the kind of ask that gets made without anyone in finance fully understanding what requires so we built migration assessment tooling that gives you complete visibility into your application estate before you move even a single workload. In the last year, we assessed over 1.1 million virtual machines. We established a base migration approach that

lets you move your own base full fallback options at every stage. In the last year, we migrated nearly 400,000 which machines, the number of VMs running on opens, grew by 417% in just the last year. Simply put clusters running VMs on OpenShift, way up customers running VMs on OpenShift, way up. Now to achieve the scale that you need, we invested deeply in the systems integrated ecosystem that will actually be in the room when this work happens. So when you think about this holistically, this isn't a migration from one VM platform to another, it's a migration really from a virtualization strategy your monetization strategy. Now, I know what some of you are thinking right now, why just go all in with hyperscaler? Just move it all to the cloud. Fair question. Here are three things that hyperscale and native services fundamentally cannot give you first consistency when you build on hyperscale and native services, you'll get a different operational model on every cloud, different tooling, security model, different networking primitives. OpenShift gives you one operational model works the same way everywhere on your private infrastructure, on AWS, on Azure, on Google Cloud, on the edge, and even in disconnected and regulated margins, second independence, when you standardize deeply on hyper scale as native services, you trade negotiating leverage For convenience. That's antithetical to the Red Hat business model. It's built on your flexibility, not your own. You start talking about open hybrid cloud 10 years ago, it is foundational to who we are. Third, regulated reality for many organizations in this room, financial services, healthcare, government, telecommunications, 100% public cloud is impossible due to legal constraints. Hybrid, then, is not really a compromise for these industries. It is the only viable strategy and Red Hat is the only platform that makes hybrid genuine, first class architecture, not now. We don't ask you to take our word for any of this. You can see what industry analysts are saying, but nothing I can say, or analysts can say is more credible than hearing it directly from the people who have actually done this work. Motorola Solutions is a perfect example of an enterprise that saw the cost of staying on legacy virtualization is not only a financial burden but a strategic risk. They saw that villain early and chose a path that prioritizes both resiliences and speed. So now let's hear about moodle's journey, directly from anuta baraku, Senior Director of core platforms and services. Hey, Emily, welcome. Welcome to Red Hat. Thanks for joining us now. Your team supports infrastructure for first responders where every second counts, destroy your infrastructure environment that you work in

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Speaker 5 25:32

Sure. Ashish, thanks for having me here at Motorola Solutions. We built that is the backbone for first responders. We're talking about dispatchers, police officers, firefighters and emergency medics, and we support them for the full incident life cycle, all the way from when the 911, call lands, to dispatching help, to responding to help, managing cases and communication throughout the incident life cycle, every second save is a second useful save a life, and we use hybrid infrastructure to provide and deploy these applications our core workflows, like call handling, dispatch this day on prem, we want To ensure near zero latency and total reliability, even when the internet fails. But we use Cloud as a vital redundancy solution and as also as a platform for advanced innovation. We deploy our AI workloads like transcription, translation, summarization, redaction, into cloud. So essentially, we are providing on prem stability with Cloud Scale intelligence. And the key caveat of all of this is that the system must be up. It must be resilient under extreme pressure. We don't just manage servers and code. We provide the certainty that the system works when the citizen calls for help. So only your team you're delivering absolutely mission critical environments and platforms. Now last year, many leaders, maybe in the room and others listening in, were hit with the virtualization cost prices face the potential 3x 5x increase in costs for your legacy footprint. Why'd you decide that red hat was a good fit with you? Yeah, I think, like many of you, we ran into this crisis, and we looked at various options, and then there were options where, like, we could just simply slap the hypervisor and live with it, but that's not what we wanted. We turned this past crisis into an opportunity to modernize our entire infrastructure, and moving to OpenShift virtualization allowed us to treat this transition as a natural evolution of our application stack. So just to give you guys some context, we managed two separate worlds. We had a legacy world of virtual machines, and we had containers running in cloud. And as you could imagine, there's a lot of silos, there's different tooling, there's different processes and operations were very inefficient. With OpenShift virtualization, we have moved to one Kubernetes page foundation that can scale across on prem and cloud and essentially unify our operations with one hybrid deployment fabric. And the transition was also seamless. We were able to run virtual machines and containers at the same time on our plan, and that allow the teams to move workloads over a period of time without having to be like a big migration project that never ends. So whether it's the cloud, the data center, the edge, we manage everything using one single lens with the same security tooling, same observatory tooling and same deployment pipelines. So

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Speaker 4 29:09

this hybrid cloud infrastructure a perfect use case for deploying open ship. Can you talk about now adopting the Red Hat platform benefiting your end users?

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Speaker 5 29:17

Yes, so we work day in day out with public safety customers. And our customer base is risk averse, and for a good reason, they need the stability on prem so that they can provide response in mission critical situations, but they also need cloud agility so that they can use these latest technological advancements to respond better, and the hybrid model has really helped us serve that purpose. Today, we provide two unique value propositions to our customers. One, we're able to provide cloud as a redundancy option, so if a call handling system on prem goes down, they're able to fall back to the cloud, or they can choose cloud as a primary and fall back to on prem. That transition is really helping. Two is like with all the advancements around AI, we are able to run our AI workflows in cloud, but seamlessly integrate them into their workflows on prem. So think of when a 911, call taker is taking a call, they're able to view live transcripts. An officer in the field is able to use translation on their radios and talk to someone in a different language. We are providing summarization across like case management and other things. So essentially, this hybrid infrastructure helped us integrate cloud and on prem in a seamless manner, and so that the end user doesn't really have to worry about where the workloads are running. That

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Speaker 4 30:47

makes a lot of sense. Only been and the platform is definitely building value for end users. In the end, this journey was about more than just a technology swap. It's operational shift for your organization. So let's look ahead at the next several years. What are the primary lessons that you take away from this modernization or inform your vision for the future model?

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Speaker 5 31:10

So I think the first step of modernization of our virtualization layer has really helped us create a strong foundation for our hybrid fabric between alternate cloud and we are able to serve everything from our core public safety systems to advanced AI models. And we are continuing down this journey. Our goal is to reduce the complexity so that our talent are working on mission critical applications, Building Better use cases, enabling better response flows without actually having to worrying about infrastructure silos. Hybrid is the end state for us. We want to take advantage of the cloud capability while maintaining absolute sovereignty of data and edge operations for our customers. So Ashish, in a nutshell, success for us means infrastructure and technology becomes invisible. It should be like oxygen, always there, always reliable and never something a police officer or a firefighter has to think about. We are not just building a platform. We are building the certainty that the tools will work when they're needed the most, in moments that matter.

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Speaker 4 32:25

Thank you already. So moments that matter indeed critical work that you and your team are doing. Your your team is really proving out that infrastructure modernization is the foundation for everything that will come next. So you didn't solve the cost problem. Didn't waste that

crisis. You built the platform for the next decade of public safety. So thank you again for sharing with us what it looks like to face downstream risks and win. Thank you for having me. What an amazing story of absolute life saving, mission critical work by an adept in the Motorola Solutions team. Now it goes without saying that the world faces uncertainty right now, from geopolitics to AI to new regulations impacting digital sovereignty, this conversation about sovereignty used to have been in the legal department, and now it's also happening in the architecture Review Board. Requirements shape how your AI models can be trained, what decisions an AI system is allowed to make autonomously, what product trade you're required to maintain every AI generated output, and that regulatory landscape is expanding. I've talked to many of you concerns about the EU, AI act, sec cyber disclosure obligations, data residence requirements across dozens of jurisdictions that continue to develop industry specific regulations, financial services, healthcare, government, all being rewritten to account for AI for us, the answer is obvious. It always has been, freedom and choice. Freedom and choice delivered by open source give you control and transparency, which is ultimately what you need, underpinning a sovereign architecture. This is the reason that the largest financial institution in the world, the largest government agencies, the most heavily regulated healthcare systems, are choosing Open Source infrastructure, the digital sovereignty solution that we've built that builds upon decades of experience, helping our customers, our partners, build on prem, private and hybrid environments any organization, whether it's an enterprise, a cloud service provider, A telco, factory, and our increasingly Neo cloud can use our platforms to build sovereign all private AI enabled clouds. Our solution spans our entire portfolio, OpenShift, REL Ansible and Red Hat AI. Today, we announced a number of new capabilities to help our customers and partners meet increasingly strict requirements from data technology and operational sovereignty. First, we've enhanced our compliance framework to make it easier to prepare for NIST two GDPR, Dora audits, compliance checks and whole lot more. We're now also offering a cross platform installer to build landing zones that immediately enforce specific operational and compliance guardrails. We built a new service fulfillment API that enables you to quickly stand up and provision service offerings and ownership. You can start with virtual machines and clusters, add new AI services like GPU or models or inferencing as a service, and as our partners, you can extend this API. We also plan to provide you in Region content delivery in your to help you safeguard your software supply chain, improve your operational resilience. You can now download rel from within the EU, and we plan to expand the CDN to additional Red Hat products by the end of this year, end of 2026 we've had Red Hat confirmed stateside support in the US for 10 years. Just in April, we launched confirmed sovereign support for the EU, ensuring support is delivered by engineers are legally authorized to work in the EU and provide support from there, and we'll look to roll this out to other regions in the near future. Help you get started as in migration for your machines. We also help release the Red Hat sovereignty readiness assessment and open source assessment that gives you a structured way to evaluate your current posture against the critical sovereignty domains, identify the gaps that you have, the ones that put your business at risk without a roadmap to close them. And finally, lastly, IBM announced a new solution, IBM sovereign core, designed to help organizations build and operate AI ready sovereign environments and verify their control, giving enterprise and governments an end to end approach to the sovereignty that's built on top of Red Hat's sovereign platforms. Now let's look at how Red Hat is already helping customers with their approach to the sovereignty by talking about something we all do, flying plane now, most of us did it this week. Really think about the GPS behind it. If your flight crossed European airspace, it was coordinated by your control, the organization responsible for managing safely over 11 million flights a year across the entire continent. Now that's a big job with a lot of moving pieces, literally. But here's the challenge, their core. It backbone is 30 years old, and with European air traffic projected to grow beyond 12 million flights annually by 2030 good enough was no longer an option. While it's robust, their infrastructure was a Walt

got made it nearly impossible to scale to meet the demands of modern aviation or to use more machine learning and AI to improve decision making. So they set out. They set out to build a unified mission critical digital platform. They didn't just want to move to the cloud. They needed digital sovereignty as a public and Pan European organization. They have to be mindful of new policies. They needed the freedom choose cloud providers while maintaining full core transparency and audit will control all the infrastructures, and this, this is where Red Hat comes in. What's really cool here is that they are refactoring 30 years of institutional knowledge into a modern API architecture, and in the process, they are transforming the post operational data analysis from a three day wait to real time capability that is already helping identify more fuel efficient flight tools across Europe for the development teams, their engineers are no longer spending their days on maintenance. They're focusing on high value applications that keep sky safe for all of us. Euro controls perfect example of how we support our customers as they face digital sovereignty challenges. They're taking a 30 year old legacy and they're turning it into fast, flexible and soft future. We are incredibly proud to be a part of that journey. Now we've spoken about the increasing regular environment and the need for operational resilience and business continuity. These are all making the case for architecture that accounts for sovereignty. We're now going to look at how that vision becomes a reality. Please join me in welcoming Raghu. Chakraborty, EVP of engineering and GM Americas for 442 Hello, Raghu, welcome Red Hat Summit. Thanks for being here with us and looking forward to hearing your thoughts about core 42 you've been established sovereign cloud operator for over five years, but this landscape has been shifting rapidly. What's the fundamental problem that you're solving for organizations in the UAE, regulated industries? Why is good enough? This is the theme that we talked about for a little while. Now no longer an option for sovereignty.

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Speaker 6 41:04

First of all, thanks Ashish for having me, and thanks for being an incredible partner for us strategic in our journey towards sovereign AI regulated industries, specifically health, defense, national security or financial services face a binary choice right now, public cloud convenience versus absolute control, autonomy and technology is key to avoiding the kill switch problem. Good enough platforms compromise sovereignty and that and that results in shadow AI or data leakage, risks that are operationally and legally unacceptable. It's certainly true Raghu about good enough platforms not being necessary. We've got to go higher than that, crossing bar exactly as a cloud provider and a NEO cloud we are trusted based layer for of the G 42 intelligence grid and the national digital platforms Across UAE, our platform enables AI applications that citizens enterprises interact with daily to operate at scale with compliance and with the reliability that critical services require. We are designed for classified, top secret workloads, extending our sovereign portfolio to the most sensitive areas of government and regulated industry. More than 70 public companies and private organizations have adopted the core 42 sovereign cloud as

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fantastic. So it's clear the core 40 is making a lot of progress here. Now we often talk about partnerships. You mentioned that as well. It's a deep engineering collaboration, and we appreciate that you were the first in the world to adopt OpenStack, 18.1 running on OpenShift for your private cloud. Congratulations to you and the team for doing that. Can you tell us about that decision to build on the stack, how our teams are working together to influence the future of certain features. While data and operational mandates were paramount, we required a platform

capable of anchoring our technological sovereignty, which is a very important component, by the way, with equal precision compared to the other requirements. Red Hat, OpenStack and OpenShift emerged as the premier solution for meeting this critical technological sovereignty requirement. By running OpenStack on OpenShift, we have unified the operational model for VMs containers across 1000s of servers that we have in production. Also, this is a really a core bill. It's all about collaboration. Our monthly cadence calls with the Red Hat engineers allows us to provide direct feedback and also, especially on must have solutions. And we have really solved about 50% of all the enhancements that we have escalated so far by using enterprise grade OpenStack and OpenShift, we provide the freedom to build and scale our own terms, on our own terms, while avoiding the foreign control or the vendor lock in that happens this collaboration turned our platform into a robust, high redundancy architecture, ensuring we meet hundreds of specific security policies set by the UAE National Security Council. That makes a lot of sense, and you really are in the business of managing national critical infrastructure, which, of course, means zero room for error. Why did he choose red hat to build this software platform? It was it came down to this right. Red Hat's work on OpenStack has turned it into a genuinely robust foundation for a full spectrum of sovereign cloud deployments, from fully air gap solutions to restricted versus connected models that we serve. The breadth combined with the stability and the maturity of the Red Hat platform is why we see red hat as a premier partner for our sovereign infrastructure. This collaboration reflects how national scale AI must be built with sovereignty, resilience and long term engineered control into its foundation. Our customers, such as large banks and government entities, trust us, because we can verify that the diagnostic data, the audit data, stays in the region, and that support is delivered by better in country personnel. This reduces the cost of complexity. We provision GPO resources for AI in minutes rather than weeks, and our national disaster recovery program for the government is already proving its ROI in terms of operational resilience. So great progress so far, undoubtedly. Let's look ahead. You've described the work in the UAE as a proof of concept for the rest of the world. So as you look to export this zombie model to other countries, how do you see the Red Hat partnership evolving to help you scale out? Yeah, that's a critical point, right? Core 42 is actually exporting the sovereign AI models, operational expertise and infrastructure frameworks that other nations can build upon. Some examples of this are the AI factory that we own and operate right here in the US, across five data centers with over 100 megawatts of AI capacity to serve the largest AI apps and the domain partnership that we announced to deploy the largest Europe's AI compute capacity, or cluster in Italy, these reflect the deliberate expansion strategy that we have around the globe. And Red Hat is our Geo expansion partner. We want to be AWS of the sovereign cloud. Imagine that AWS for the sovereign cloud next, decade of AI will separate the organizations that build resilient, durable foundations from those that scale quickly on fragile ones. Whether it's connected, as I said, restricted or fully air gapped, we are building the standard global platform. Partnering with Red Hat gives us a common technology platform that lets us modernize digital infrastructure anywhere in the world.

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Speaker 4 48:29

Well, love that. Thank you for sharing your vision with us. You're obviously proving that at 442 you don't have to trade choice or control, can certainly have both. So thanks again for joining us. Thank you. What a great story. Love the vision, love the ambition of core 42 and Raghu and team. Now data sovereignty is the foundation, but AI sovereignty, as you heard from Raghu, is the next evolution of the same challenge, and it is where the stakes are significantly higher. Let me define AI sovereignty precisely the thing. It's a term that's getting used loosely. There are three things you must own to have genuine AI sovereignty in your organization. One model, sovereignty, the ability to choose which AI models run your environment, the ability to change them as the market evolves, we're all seeing it will evolve dramatically, the ability to run them anywhere based on what your business, your data, your regulators required two data sovereignty, the peace of mind that the data you use train, fine tune and run your AI models never leaves your control, that it stays in the jurisdictions your regulators require that it's not used to train Anyone else's models without your explicit consent, that you know, that you know precisely and verifiably where it lives and who has access to it at every moment, three outcomes, the ability to audit every significant AI generated decision to explain it, trace it back to the data and model that produced it, and if necessary, to override it. AI sovereignty isn't a theoretical conversation. Customers are actively addressing this today. Caladon was founded in 1855 and began its journey as a state operated telegraph service in Norway for 170 years, right? 170 years, Telenor has stayed at the forefront of every major communication ship today. They are one of the most geographically diversified communications groups in the world, serving millions across the Nordics as a nation. Here's a telephone giant that's operating in highly regulated markets. Their challenge is the same villain that many of you face operationalizing AI at scale while navigating a landscape of regulatory and security constraints. Tell most customers, the government agencies, healthcare providers, critical infrastructure operators that need AI that respects national borders they need high performance infrastructure for model training and inference that remains verifiably secure and under the control. They need sovereign AI, the telephone story bridges sovereignty and AI, it's a great example of how closely these domains intersect for our customers globally. Telenor did not wait for the market to define its future. They built the Telenor AI factory. The foundation of that is the Red Hat AI factor, Nvidia. You'll hear more about that later in this keynote. This is Norway's first sovereign and secure AI cloud service. They're already hosting multi tenant customers in Norway. They're hosting Norway's own language models, and they're accelerating industrial automation, voice to voice translation, private 5g network optimization, just as some example customer use cases. So having heard all that Telenor is just not HELCO anymore, they're sovereign, AI provider. So if an organization can margins over 170 years of history, and if an organization can grow from managing telegraphs to being a sovereign AI powerhouse, that every organization in this room and every one of you listening to us today has the potential to do the same. Now, let me bring on our CTO. Chris Wright, onto the stage. We'll go deeper in AI for the rest of the keynote.

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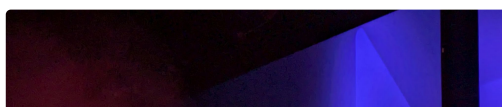
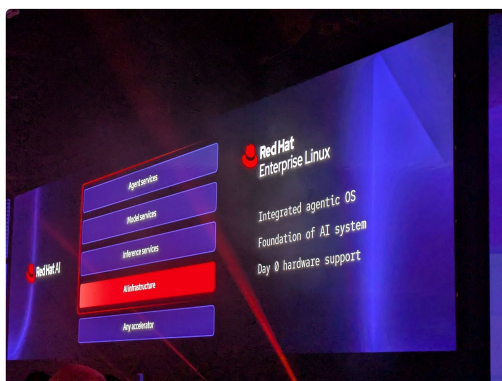
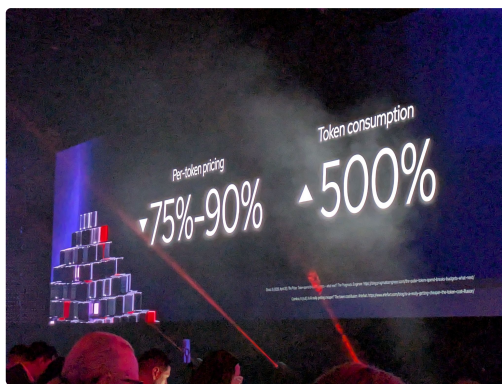
Please welcome Red Hat Chief Technology Officer and Senior Vice President of Global Engineering. Chris Wright, you

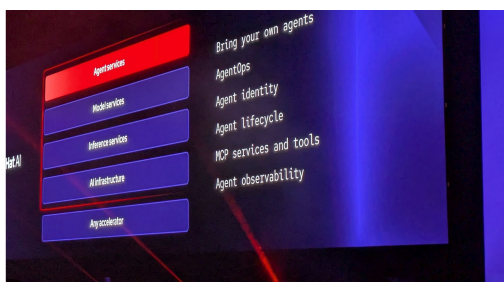


I tell the graph to AI man, I love telenor's modernization story because they didn't just wait for someone else to define their future. I mean, sitting out on the AI sidelines, while tempting given all the change, is a totally failed strategy. In my 20 years at Red Hat, I just never seen any technology move at this pace. If you're overwhelmed by the number of new things happening in AI, I can say with confidence, you're not alone. It feels like there's hundreds of new announcements every month, sometimes daily, but let me show you three examples of how I think about the speed of change happening across AI. Three and a half years ago, everybody had that. Do you remember where you were moment when generative AI and chat GPT went from more of a research concept to something that millions of people were just using immediately? Nobody was prepared for this level of change, the least of which was enterprise, IT organizations. Then, eight months later, meta responded with an equivalent open weights model for mass adoption that was llama two. Two years ago came reasoning, reasoning models like open ai 01 could now think longer about problems. Reasoning creates this massive explosion in token usage, and it's laid the groundwork for future AI agents, and then just five months later, deep sea car one, an equivalent reasoning model was released under open source license at a much lower cost to train. Now we've had quad code since February last year, but with the introduction of Opus four, five toward the end of last year, in November, we saw this emergence of an agentic coding tool. I mean, everybody came back from the holidays with their minds blown about their agentic coding experience. So that combo in late November, just a few months later, two months later, openclaw became the fastest growing open source project ever, like open source agents for everyone. So we had eight months, we had five months, we had two months. Every time open responds faster than the time before, admittedly, three years ago, it wasn't totally clear about the role of open source with AI that's critical for deep dependencies in data science, but now with open source inference, with open weights, models with open source harnesses, these are undeniably ready to be the open foundation of AI in the future. So I'd recommend take away two things from these examples. The first is this speed and velocity is not going to slow down anytime soon, so you got to make sure you're building flexibility into your AI strategy, and the second, the right AI platform, built on open source, will enable you to manage the pace of choice and change that AI is driving today. Now, before we dive into the platform, let's just reflect on the business reality that makes this platform so important. Right now, most enterprise AI runs like this. You call a frontier model API. You pay for the tokens you send it and the tokens that it generates for you. It's super easy to start powerful models and for some workloads, genuinely just the right answer. But here's the dynamic that's worth watching closely, per token pricing is coming down 75 to 90% a year. That's the good news. Token consumption, however, is skyrocketing, increasing over 500% a year, in some cases, and so with these new reasoning models that are generating even more tokens as they think consuming 10 to 20 times more tokens than standard models. And with AI agents, you'll have to expect another 5x multiple because an AI agent reasons it plans. You heard from Matt. It calls tools. It iterates, it loops, it's connected to other agents in a workflow. So to win token economics, you have to move from being just a token consumer to an actual token provider. And we're working with organizations around the world to enable just that. Organizations that own part of their own inference infrastructure, that run self hosted models, optimize for their business, and have real choice over which model to use for which job. Those are the organizations that will emerge as the winners. All right, so let's dive into the platform. Now, remember what Matt described earlier, that journey from Frontier models to open source models hosted internally. Matt ran on this platform, Red Hat, AI enterprise, it's the platform that really made it possible. We literally call this metal to agents, a fully integrated stack that spans from the physical machines running AI accelerators all the way up to the agents delivering real business value every layer connected, secured and open at the foundation Red Hat AI enterprise runs on any accelerator and any cloud, giving our customers choice and control over

where they deliver AI capabilities for their business. This includes accelerators from Nvidia, AMD, Intel, and from the cloud providers, custom silicon, like in AWS, in Google Cloud and Azure, and whatever comes next across public, private and edge environments. That's the foundation. Now let's go through the Red Hat Enterprise AI stack that makes the metal to agents possible. It begins at this AI infrastructure layer. Now this is where the IT operations, SRE and platform engineering teams live today. AI runs on Linux and is scaled with Kubernetes. Red Hat provides this Linux Foundation that trusted and reliable operating system that's built with the security needs of both current and future workloads in mind, so we protect the foundation of your AI system. Everything above this really depends on getting this layer right. Red Hat, OpenShift is our unified Kubernetes platform for virtual machines, for containers and for AI workloads, network isolation can control what an agent can reach, and GPU sharing across workloads means your most expensive resources are not locked to a single team, or worse, just sitting idle between inference calls. Next are inference services. This is where your token economics layer lives, where performance, where scalability, efficiency and cost. This is where it all comes together. Inference is the layer that most directly determines whether your AI strategy succeeds at scale. If you get inference wrong, your customers aren't happy, your data science teams aren't happy, your AI agents aren't happy, and your CFO is definitely not happy. So we deliver world class inference services through bllm, the open standard for inference serving. And Red Hat is super proud to be the leading contributor to the bllm project. Now bllm is complemented by LLMD, our open source distributed inference framework. We built LLMD because AI workloads aren't like typical cloud native workloads. Enterprise AI workloads often don't fit on just a single GPU. They're stateful. They fundamentally rely on KV cache, and they vary dramatically from request to request. So LLMD intelligently routes and schedules requests across your environment, optimizing how models execute, how GPU resources are used, and how latency is managed. Remember those token economics I mentioned, in just one year, the performance improvements with LLMD are three times more output tokens and 10x faster time to first token. And there's a great paper that was recently published by Red Hat and Tesla, along with the open source community that has all the details. Next up, we've got model services. Now this is where choice is critical, given all the changes happening with Frontier models, with open weight models, with domain specific models and even small language models. Now the starting point for many companies is delivering models as a service to their internal application and data science teams. Red Hat AI enterprise just makes that easy. When your AI teams need to add, remove or swap a model, you do it in just one place. Your teams choose the models they want, token quotas, credentialed access, request priorities and rate limiting. All of it is delivered through our AI gateway. It's one control plane for every model interaction across your environment. Now the reality is that companies will use many different models to deliver their AI capabilities that their business needs, and our vllm team has deep expertise with a wide variety of models. In fact, almost every major new model update is optimized for vllm Right on day zero. So to bring all that innovation and optimization to our customers, the validated models by Red Hat program has more than 20 collections of models and nearly 700 variations of models that have all been validated on bllm and then optimized for speed and efficiency. So this should give you a sense of our commitment when we say that Red Hat AI supports any model. In order to ensure that you can bring your company's data to your models, we've integrated a number of data services to simplify owning your AI experience things like retrieval, augmented generation or rag and fine tuning, and, most importantly, integrated evaluations. All right now we're at the top agent services. Now. This is where AI will make the biggest impact on differentiating your business into the future. Agents are no longer just an experiment for a clever automation they are core to every enterprise AI strategy. It won't be unusual in the future to hear that a company has 1000s or 10s of 1000s of agents. So our goal is to make sure you have the AI platform that can manage those agents securely today. Fact, I'm 100% certain that there's some AI agents running somewhere in your

organization right now, and they bring with them a series of challenges. The first challenge is the exponential impact that AI agents will have on your inference requirements. Remember, if you don't have control over your inference usage, you don't have control over your token economics. The second is sprawl your developers want Lang chain your data science team is building with crew AI, your marketing director figured out open claw, and hundreds of your employees just downloaded quadcode or Codex. So if your AI strategy doesn't manage sprawl and give your team's choice, you're going to have frustrated employees. You frustrated employees. And if your AI agent security strategy doesn't support multiple types of agents, you're going to have more than frustration from your security and compliance teams. So remember Red Hat. AI supports any cloud, any accelerator, any model and any AI agent, we call this, bring your own agents to enable this level of choice. The AI platform needs to deliver agent ops. Identity matters. Every agent gets a verified, authorized identity. Lifecycle matters. Every agent can be versioned, updated and rolled back. Tool calling matters so native support for MCP services connects agents to the tools and data that literally define your enterprise. And of course, most importantly, tracing and monitoring matter integrated observability enables the platform to track agent interactions. So when you give your team's choice and your IT teams control, this turns bring your own agents into agents as a service that's managed by the AI platform.





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to bring it all together, Red Hat delivers the complete enterprise AI stack from the AI accelerators up through the AI agents. It's tightly integrated. It's totally open with our ecosystem partners, with every layer connected, secure and open. So now let's check out how this goes into action with BNP Paragon, with a footprint in 64 countries and territories and more than 180,000 employees BNP is a leader in banking and financial services across Europe. Now, AI has been a priority since 2016 and they view it as a strategic asset for developing a new way to bank. Now, most companies talk about AI in terms of saving money or being more efficient, but what about AI leading to value and growth? So far, the bank says it's generated nearly \$600 million in additional value from ai, \$600 million in additional value from leveraging the strategic asset. Now their IT organization relies on some 750 AI experts and 3000 data experts, and today, BNP is architecting the next era of finance by scaling one of Europe's most formidable AI factories, powered by Red Hat AI to fuel 1000 AI use cases, the bank has unified its predictive and generative AI strategy across a hybrid cloud architecture, and that AI factory is helping VFP industrialize AI use cases and deploy generative AI models at scale. So the AI factory platform leverages GPU computing to operationalize AI capabilities across the bank, and they've transformed GPU provisioning from like this weeks long bottleneck into just a minute's service, and they've also accelerated their internal LLM as a service platform, while upholding the highest standards for digital sovereignty. So BNP Paribas has made tremendous strides in operationalizing AI for its business users, and let's not forget about that \$600 million in additional value that I mentioned. That's something that any board would love to hear about these days. So what we're doing today for AI is what we did for all the other pillars of enterprise. It think Linux, Kubernetes and Ansible. And we're doing this in collaboration with our partner ecosystem, including Nvidia, where we've been heads down co engineering, a foundation that helps you move past the science project phase and into actual production. So joining me now to talk about how we're accelerating that path is Justin Boyton,

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all right. DJ, JV, right here.

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So Justin, I gotta say, our joint work on the Red Hat AI factory with Nvidia has been like a highlight for the year, and from my perspective, that's about giving customers this consistent and sovereign foundation across the hybrid cloud so that they can own their own intelligence. But what are you seeing in terms of interest from customers about this integrated



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approach? Well, first, it's awesome to be here with you, Chris, and it's great to be back at Red Hat Summit. Look at this audience. Open source is born, right? Red Hat is the company that is synonymous with industrializing open source software for enterprise. And every time we talk to enterprises, they consistently tell us they need a platform to build, deploy and scale specialized AI agents that understand their business. They can automate critical workflows with them, and they can run them securely across a multi cloud and hybrid cloud environment, and so this Red Hat AI factory with Nvidia optimizations brings us that foundation for every enterprise. It combines Red Hat's open source platform with Nvidia accelerated run times our open models, our open inference stack, so that customers can industrialize AI agent development and deployment with really the best token and power efficiency.

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Yeah, I think that token power efficiency fits right into the token economics we were just talking about. So Red Hat AI's upcoming 3.4 release includes the new governed model as a service experience that's delivered via the Red Hat AI gateway that gives developers that seamless access to created models like Nemo Tron through the standard open AI compatible interface. We've already announced GA support for Nvidia, Grace Blackwell and the roadmap to Land Day Zero support for the upcoming Vera Rubin architecture. Now in the past, software, surprisingly, could lag behind hardware for months, if not even longer. You You guys are moving at the speed of light. Why is this tight synchronization between the Red Hat stack and Nvidia's latest silicon so critical right now? Yeah,

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I think the only in your setup of all of this, only point you may have missed is we are living through the world's biggest industrial revolution as we build out AI. And when we think of the infrastructure being deployed in AI, it's really what we call a five layer cake. There's energy, there's chips, there's the infrastructure, software infrastructure, you talked about models and applications of the agents, and you've got to fully optimize all of that so that you get the best token efficiency to automate work. And really that's the name of the game, is you got to convert power into tokens, and these tokens are going to be the intelligence that powers your business. When we build a new rack scale architecture like our Grace Blackwell system, we're building a system with six processors. We go through a process we call extreme co design. And in this extreme co design process, we make sure we squeeze every bit of power efficiency out of that rack. But to realize the benefits, you've got to continue to optimize the software infrastructure, the model serving layer and the agents, so that when customers set up these new platforms, they immediately get their biggest business advantage on these newest architectures.

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Yeah, you can see the importance your extreme co design at the chip level, and that connection with software super important. Now we've already talked today about that moving from simple chat bots to fully autonomous agentic AI, super exciting, but let's be honest, it definitely brings up some pretty serious security concerns. Now we're collaborating on the open shell project to bring sandbox, zero trust environments to these agents. How do you see this open shells project and kind of the security first mindset helping enterprises to trust that agents can actually do these long running tasks and perform real, real value and potentially regulated environments.

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Yeah. And so one of the big advancements you mentioned the open cloud moment, and agents are advancing at this tremendous speed, one of the fundamental innovations of open cloud was self evolving, so it was trying to write software on the fly and reinsert it into the application, which can be, you know, absolutely terrifying. And so where we are as an industry as we believe that's a huge productivity gain if you can deploy it in the right way. We think every company is going to deploy agents this year to every employee. And how do I know this? And why am I so confident? Well, we've done this at Nvidia. We gave all 40,000 employees access to the first agent, and now employees want more. But as you start to deploy these agents into every workflow for chip design, software development, supply chain, automation into like how we run the business across the board. You know, agents become core to our work. We want to run them and the intelligence behind them as much as we can on infrastructure that we operate and we control, so that we make sure we can control our costs. But you got to do it in a way that you can govern it, and you can govern these agents. So just like we give our employees role based access control into all the business systems and tools that they use to do their job. We've got the same thing with agents. And so first we built this platform we call open shell. It's a secure agent governance layer that provides fine grade policy controls to govern these agents at scale. And then the second thing we did is we're introducing confidential computing into the Red Hat AI factory with Nvidia together. And we're doing this so that you guys can ask of your Frontier model companies to run on infrastructure you own. And we're starting to see, actually, quite a bit of excitement around that as well. So one thing I think, from Nvidia's experience, which is pretty clear, is that no matter what agents people want to use, to your point, whether you're using cloud or Codex, Lang chain, building their own agents, we want to have kind of a unified, secure runtime to be able to govern them at scale. And then we see the world being a multi agent, multi cloud, hybrid cloud world across every enterprise globally.

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Yeah, I love that,

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multi agent, multi cloud, hybrid cloud.

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So from my point of view, I've often thought that for AI to really work, organizations need technical independence. They can't be locked into just a single black box. So think about that. How our work on these open source runtimes and that consistent infrastructure? How does that help a CTO feel confident that they're not just renting their future but actually building a sustainable, independent AI strategy? Yeah, and I'll say, Chris, I couldn't agree more. A lot of this comes down

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to like you have to control your destiny in this transition. And so having choice on where you run the agents, how you control the context, which is all the information about your business going to these agents, is absolutely critical, and our collaboration being built on openness is a key part of this. So by combining really Red House leadership and open source software with open models from Nvidia, open data sets that we provide the industry so people can post train these models, open post training techniques, open agent optimization techniques and open shell as the runtime customers have the choice. They can run anywhere they want with really commercial support offered together between Red Hat and Nvidia, we believe we're now giving enterprises really the keys to, you know, owning their their ultimate outcome with agents?

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Yeah, I love that, giving the agency to our to our customers and keys owning their own AI strategy. I mean, you can see there's a ton going on, and clearly a lot more work to do this. Co engineering has been really exciting, and we're doing tons of great work and engaging directly with customers. But for now, Justin, I got to say just thanks for being here. Thanks, Chris, it's awesome. Thank you guys. All right. Well, Verizon is the perfect customer story to close the morning, because this us, telecommunications giant has operationalized so much of what we've been talking about today, from virtualization to scaling. AI, they're proving that a unified open platform is the only way to modernize a footprint as big as theirs, while staying scrappy and reliable. So please join me in welcoming Praveen atrey to the stage. I love your CEO has been, like, super vocal about this Verizon becoming a scrappier competitor through radical customer simplification. So now, from an engineering standpoint, when you're managing this massive scale network like yours, how does moving to a unified platform actually deliver that simplicity while maintain the urgency needed to stay ahead of this competitive market. Chris, thank

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you for having me here. Good morning. Everyone. Great to see you all at Verizon, we've embraced that in order to thrive in a hyper competitive market such as ours, you need radical customer centricism, and that requires simplicity, speed and efficiency. What that demands from a network is a software centric network model, essentially a network that moves at the speed of software and the key enabler for that is a powerful and a performant private cloud platform, not just at the core in the data centers, but also in the edge, the far edge of the network as well. And this is what offers control, security, observability, observability and a host of common capability that essentially enables every network element to now become an app, an application. And what this does is that it allows us to fully embrace the software and the cloud operating model. This is what drives continuous improvement through seamless upgradability, self healing and continuous self optimization, which in turn delivers more value, more delight to our customers over time. I love that

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network software, just an application, you make it sound so easy. Now the Verizon cloud platform has been on like, a 10 year journey, and while G is cloud native, virtualization is still totally a part of your reality. So I know you've been working on consolidating legacy environments all along the way. What's the strategic value for Verizon in having a single platform that manages your containers and your virtual machines,

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rather than separating them into two separate worlds? Thank you, Chris for the mention on the Verizon cloud platform, or VCP, as we call it internally. It's a 10 year anniversary. You've been a strong partner throughout the journey from Day Zero. Thank you for that. To answer your question, when you run a network and a business at the scale of Verizon, and I'll throw out a couple of stats with nearly 150 million retail wireless connections. That's just mobile. In addition, we also have nearly 17 mil broadband connections. I could go on and on with the vanity stats, but I want but a natural consequence of this scale is that you also end up with heterogeneous and a multi generation technology and an application landscape you're simultaneously running modern, highly decomposed micro services alongside legacy, monolithic applications or VMs, and this is where having a Single hybrid platform that handles all of this seamless transition of workloads, enabling various heritages and applications to work closely together. It makes a big difference. It makes a big difference towards simplicity and efficiency and again, ultimately enables us to be better for our customers. Yeah, I like the picturing heirloom applications. We got some, some interesting legacy there. Yep. Now we've been talking about AI this morning, and it's definitely moving from that, that buzzword, to a core operational tool for enterprises. And I know you've described the goal of building autonomous networks. I mean, it's basically like a self driving network. So how is this kind of CO developed platform that we're creating

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with OpenShift AI, helping you bridge the gap between just collecting data and actually automating the network in real time to minimize any customer disruption,

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this is a great one, the data, the insight, the recommendation to action loop has been poor and happens repeatedly in the context of network automation today, in fact, not just for plan changes like lifecycle management and upgrades for unplanned, rainy day fault remediation and for performance improvements. We have made millions of changes in 2020 25 alone in closed loop. More no human in the loop. And this is fascinating in its own right. But while this is true, where we're going is 10x with AI, everyone is a developer, and with agentic AI, your best thoughts can be amplified and scaled almost limitlessly. And so what we can do in the context of agenda AI with network automation is is far beyond what we can do with structured automation. For example, a network operator could detect network experience degradations for each individual customer rather than aggregate alarms that smooth out individual customer related data points, and then you can drive complex triage loops for each incident with the reasoning capabilities these agents offer, which can then autonomously drive recommendations that lead to actions that resolve these customer issues in near real time before they ask for help. And while that is super powerful, that's just one of the scenarios that AI in network enabled through AI runtimes such as OpenShift AI. And of course, the hyper evolving models of today can help in terms of building, planning, running the network that certainly demonstrates radical customer centricity, like a personalized network. Just for me, I appreciate that. How about just given the work that we've done? You know, what are your experiences in realizing just benefits from this whole modernization journey that you've been on? Yeah, there are many you can say this better than I can in many ways, but I will say, let me name a three or a couple. First of all, we've moved on from the state of forced hardware upgrade cycles. We have a complex network with multiple vendors, and we use to have situations such as, hey, if you want that feature, you need a new box in order to support that. We don't have that anymore. For elastic, we can add hardware horizontally. In addition, the modularity of software is another key benefit. This enables us now to replace components, such as replacing a stochastic algorithm for channel state estimation with a AI deep neural network algo that does the same job, but.